

JavaScript Array Method - join()

The `join()` method is one of the most useful array methods in JavaScript. It is used to **convert an array into a string** by joining all elements together.

You can choose any separator such as a comma, space, dash, or any custom character.

Unlike `splice()` or `push()`, the `join()` method **does not modify the original array**.

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What is join()?

Definition

The `join()` method combines all elements of an array into a **single string**.

The elements are separated by a specified separator.

If no separator is provided, JavaScript uses a comma (`,`).

Why Use join()?

We use `join()` when we need to:

- Convert an array into a string
- Display lists
- Generate CSV data
- Create URLs
- Format user input
- Build readable text

Syntax

```
array.join();  
  
array.join(separator);
```

Parameters

Parameter	Description
separator	Optional. Character(s) placed between elements

Common separators

```
","  
" "  
"_"  
"|"  
"/"
```

Return Value

Returns a **string**.

The original array is **not modified**.

How join() Works

Array

```
["Apple", "Mango", "Orange"]
```

↓

```
join()
```

↓

Result

```
"Apple,Mango,Orange"
```

Array

```
["Apple", "Mango", "Orange"]
```

↓

```
join(" - ")
```

↓

Result

```
"Apple - Mango - Orange"
```

Example 1 (Default Separator)

```
const fruits = [  
  "Apple",  
  "Mango",  
  "Orange"  
];  
  
console.log(fruits.join());
```

Output

```
Apple,Mango,Orange
```

Explanation

Since no separator is provided, JavaScript uses a comma (,).

Example 2 (Space Separator)

```
const words = [  
  "I",  
  "Love",  
  "JavaScript"
```

```
];  
  
console.log(words.join(" "));
```

Output

```
I Love JavaScript
```

Example 3 (Dash Separator)

```
const numbers = [  
  10,  
  20,  
  30  
];  
  
console.log(numbers.join("-"));
```

Output

```
10-20-30
```

Example 4 (Pipe Separator)

```
const colors = [  
  "Red",
```

```
    "Green",  
  
    "Blue"  
];  
  
console.log(colors.join(" | "));
```

Output

```
Red | Green | Blue
```

Example 5 (Empty String)

```
const letters = [  
  
    "J",  
  
    "S"  
];  
  
console.log(letters.join(""));
```

Output

```
JS
```

Explanation

No separator is added between the elements.

Example 6 (Single Element)

```
const items = [  
  "Laptop"  
];  
  
console.log(items.join());
```

Output

```
Laptop
```

Example 7 (Empty Array)

```
const data = [];  
  
console.log(data.join(", "));
```

Output

An empty string is returned.

Example 8 (Original Array)

```
const fruits = [  
  "Apple",  
  "Mango"
```

```
];  
  
const result = fruits.join(" - ");  
  
console.log(result);  
  
console.log(fruits);
```

Output

```
Apple - Mango  
  
["Apple","Mango"]
```

The original array remains unchanged.

Real Project Example 1

Display Student Names

```
const students = [  
  "Rahim",  
  "Karim",  
  "Rokon"  
];  
  
console.log(students.join(", "));
```

Output

```
Rahim, Karim, Rokon
```

Real Project Example 2

Generate URL

```
const path = [  
  "users",  
  "profile",  
  "1"  
];  
  
console.log(path.join("/"));
```

Output

```
users/profile/1
```

Real Project Example 3

CSV File

```
const row = [  
  "John",  
  25,
```

```
    "Bangladesh"  
  
  ];  
  
  console.log(row.join(","));
```

Output

```
John,25,Bangladesh
```

Real Project Example 4

Tags

```
const tags = [  
  "JavaScript",  
  "React",  
  "Node.js"  
];  
  
console.log(tags.join(" | "));
```

Output

```
JavaScript | React | Node.js
```

Real Project Example 5

OTP Code

```
const otp = [  
  4,  
  8,  
  2,  
  9  
];  
  
console.log(otp.join(""));
```

Output

```
4829
```

Best Practices

Use `join()` when converting an array into a readable string.

Choose an appropriate separator.

```
array.join(", ");  
  
array.join(" ");  
  
array.join("-");
```

Store the returned string.

```
const text = array.join(", ");
```

Use meaningful variable names.

```
studentList
```

```
csvData
```

```
urlPath
```

```
otpCode
```

Common Mistakes

Mistake 1

Thinking `join()` returns an array.

Wrong

```
const result = fruits.join();  
  
console.log(typeof result);
```

Output

```
string
```

Mistake 2

Expecting `join()` to modify the original array.

```
const arr = [1,2,3];  
  
arr.join("-");  
  
console.log(arr);
```

Output

```
[1,2,3]
```

The original array remains unchanged.

Mistake 3

Forgetting the separator.

```
array.join();
```

Output

```
Apple,Mango,Orange
```

JavaScript uses a comma by default.

Mistake 4

Confusing `join()` with `concat()` .

```
join()
```

Returns a **string**.

```
concat()
```

Returns a **new array**.

Interview Questions

What does `join()` do?

It converts an array into a string.

Does `join()` modify the original array?

No.

What does `join()` return?

A string.

What is the default separator?

A comma

```
","
```

Can we use a custom separator?

Yes.

```
array.join(" | ");
```

What does `join("")` do?

It joins all elements without any separator.

Which method combines arrays?

```
concat()
```

Summary Table

Feature	Description
Method	<code>join()</code>
Purpose	Convert an array into a string
Parameters	Separator (optional)
Returns	String
Changes Original Array	No
Default Separator	Comma (<code>,</code>)
Common Uses	CSV, URLs, OTP, Display Lists

Quick Comparison

Method	Returns	Changes Original Array
<code>join()</code>	String	No
<code>concat()</code>	New Array	No
<code>push()</code>	New Length	Yes

Final Notes

The `join()` method is one of the easiest ways to convert an array into a formatted string. It is commonly used in:

- CSV file generation
- URL creation
- Displaying names
- OTP generation
- Tags and categories
- Report generation

Mastering `join()` will help you format array data efficiently and prepare you for working with strings, APIs, and file generation in JavaScript.